MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 11 (1-5 Nov 2026) - Period 1	Unit 3 - 3.2 Styling the Game Page

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 11 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Game pages complete for ~85%; gallery homework shows good layouts. Today: CSS Flexbox, then style the index and game pages in one period.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students use CSS Flexbox (display:flex, justify-content, align-items, gap, flex-wrap) to lay out and style the index page and the game page (glove area, holes, score). Sprint 3.2 obj. 1-4; CSTA 3A-AP-16, 3A-AP-21.

Success criteria — *what will successful learning look / sound like?*

I link style.css; I use at least 4 flex properties correctly; my glove area highlights the selected glove; my game page works at laptop and phone widths.

Key knowledge / skill / understanding:

Selectors target elements; Flexbox arranges items on a main and cross axis; classes reuse styles; flex-wrap helps responsive layout.

Likely misconception / access or challenge need:

justify-content and align-items are the same; flex goes on the child; fixed pixel widths are responsive.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*	ACTIVITY 2 *student task + evidence of learning*	ACTIVITY 3 *student task + evidence of learning*
Strategy: Gamification **Time:** 8 min	**Strategy:** Project-based task **Time:** 23 min	**Strategy:** Cooperative learning **Time:** 6 min
What students will do: Flexbox challenge: teams predict the flex rule that moves projected items into place, then test it live.	**What students will do:** Style the project: index page centred with flex; glove customisation area with a .selected state; game page holes grid, score board and tray using the project CSS guide.	**What students will do:** Pairs test each other's pages in DevTools device mode and record one layout issue + fix.
Evidence tool: Mini-whiteboards Assessment: ☑ Diagnostic ☐ Formative ☐ End Grouping: ☐ Individual ☐ Pair ☑ Group	**Evidence tool:** Rubric Assessment: ☐ Diagnostic ☑ Formative ☐ End Grouping: ☑ Individual ☐ Pair ☐ Group	**Evidence tool:** Peer-assessment checklist Assessment: ☐ Diagnostic ☑ Formative ☐ End Grouping: ☐ Individual ☑ Pair ☐ Group
Resources: Flexbox practice page; display board.	**Resources:** VS Code; QR 'Flex properties', 'Design the glove customization area'; project CSS sources.	**Resources:** Browser DevTools; checklist.
Support (low performers / SEN): Property card with arrows for the two axes.	**Support (low performers / SEN):** Starter CSS with blanks to fill (justify-content: ___).	**Support (low performers / SEN):** Checklist with pictures of good/bad layout.
Stretch (fast learners / gifted): Combine 3 properties in one answer.	**Stretch (fast learners / gifted):** Add a media query for under 600 px and CSS variables for a colour palette.	**Stretch (fast learners / gifted):** Suggest the exact CSS property to change.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: the same page without and with CSS. Link the stylesheet live. Draw main and cross axes on the board; model display:flex, justify-content, align-items, flex-wrap with instant refresh. Show the .selected class for gloves. Share rubric.

What is new or innovative in this lesson? *(ETEC looks for this)*

Students learn flex through a team game, then apply it straight to their own game, testing responsiveness with DevTools.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Stylesheet path wrong (no styling); flex applied to the child not the parent; images not showing (wrong path); spending all the time on colours. Likely: 'nothing moves'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket: given a layout picture, write the 3 flex properties needed; name one responsive technique used. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: finish styling; upload screenshots at laptop and phone widths. Read student book on event listeners.

5 REFLECT — After the lesson *Complete within a day. This is reflective-practice evidence and feeds your next Starting point.*

What worked — and how do I know?

Gamified start worked; most applied flex correctly after the axis drawing. Several students used colours from their national flags thoughtfully.

What will I change next time?

Two sprints' styling in one period is tight: set a 12-minute checkpoint for the index page before moving to the game page.

Students needing follow-up:

Students with broken image paths (approx. 3 per class) and SEN students need a 1-page CSS reference.

COVERAGE CHECK *Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.*

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
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Create something new · 2-1-1-7	**Discussion** expressing ideas · 2-1-1-8	**Pair / group** cooperative work · 2-1-1-8	**Learner choice** independent work · 3-2-1-6	**Self-assessment** 2-2-1-1	**Peer assessment** 2-2-1-1	**SEN / low** specific support · 2-1-1-3	**Gifted** specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☐2 ☑3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☑C ☐H	☐1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback

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Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Studies (DigiChamps L10)	Grade 10 - Classes 10/1 to 10/6	Wk 11 (1-5 Nov 2026) - Period 2	Unit 4 - 4.1 Setting the Game Elements (1/2)

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 11 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Styled pages uploaded by ~80%; image paths the main issue. Today JavaScript begins: event listeners to make the index page respond.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students script the index page with event listeners (click, input), use the DOM to read player name/glove choice and pass data to the game page with sessionStorage. Sprint 4.1 obj. 1-2; CSTA 3A-AP-16; Challenge 1.

Success criteria — *what will successful learning look / sound like?*

I link script.js; I add an event listener to a button; I read and store the player name and glove in sessionStorage and read it on the game page.

Key knowledge / skill / understanding:

DOM = page as a tree; events trigger functions; sessionStorage keeps data for the session.

Likely misconception / access or challenge need:

sessionStorage keeps data forever; onclick in HTML and addEventListener are unrelated.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 10 min

What students will do:

Watch-and-try: select an element with getElementById, add a click listener that changes text; console.log values in DevTools.

Evidence tool: Mini-whiteboards

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

VS Code; browser console; QR 'Event listeners'.

Support (low performers / SEN):

Code template with blanks; pair with a buddy.

Stretch (fast learners / gifted):

Use querySelectorAll to add listeners to all gloves.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Problem solving **Time:** 20 min

What students will do:

Script index page: read player name, store chosen glove, validate empty name, and on Start save to sessionStorage and open game.html; on game.html display the name.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Project source 'Communicate between the web pages'; student book 4.1.

Support (low performers / SEN):

Step-by-step flowchart and starter code.

Stretch (fast learners / gifted):

Challenge 1: 6x5 interactive grid highlighting clicked cell.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Think-pair-share **Time:** 7 min

What students will do:

Real-life link: where do websites remember you (cart, language choice)? Pairs compare sessionStorage vs cookies/localStorage and privacy concerns.

Evidence tool: Exit ticket

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Slide with examples; link to Unit 1 Data Privacy.

Support (low performers / SEN):

Picture examples of online shopping cart.

Stretch (fast learners / gifted):

Explain privacy risks of storing personal data.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

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My setup / modelling: Time: 6 min

Hook: click 'Start' on the finished game - how does page 2 know my name? Model DOM tree on the board, event listener syntax, console.log debugging, sessionStorage.setItem/getItem. Key words: event, DOM, propagation, session.

What is new or innovative in this lesson? *(ETEC looks for this)*

Debugging routine with DevTools console taught explicitly; links JavaScript storage to data-privacy ethics from Unit 1.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☐ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Script tag in wrong place (element null); typos in ids; using = instead of ===; forgetting to check Console for errors. Likely: 'nothing happens when I click'.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket (Forms): order 4 lines of an event listener; what sessionStorage does. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Madrasati: finish index script; Challenge 1 grid (optional). Watch QR video 'Loops in JavaScript'.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Console debugging helped: students read errors instead of asking straight away. Privacy discussion drew on different cultural views of sharing data.

What will I change next time?

Model the script placement (end of body / defer) first - most 'null' errors came from that. More pair programming for beginners.

Students needing follow-up:

Approx. 4 per class still cannot pass data between pages; small-group help next lesson. Gifted: grid challenge done by 6 students.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy read + write · 2-1-1-6	Numeracy numbers / data · 2-1-1-6	Digital students use it · 2-1-1-4	Digital research trusted sources · 2-1-1-4	Real-life ① application · 2-1-1-5	Real-life ② application · 2-1-1-5	Inquiry experiment / research · 2-1-1-7	Thinking problem-solving · 2-1-1-7
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